

TAGUM GLOBAL MEDICAL CENTER INC

Radiology Department



TGMCI RADIOLOGY QUEUEING SYSTEM

System Documentation

Requirements | Design | Testing | User Manual

Prepared for the Radiology Department

Tagum Global Medical Center Inc

September 17, 2026

TABLE OF CONTENTS

A. BUSINESS REQUIREMENTS	1
B. USER REQUIREMENTS	3
C. SYSTEM REQUIREMENTS SPECIFICATION	5
D. ACTIVITY DIAGRAM	8
E. USE CASE DIAGRAM	10
F. CONTEXT FLOW DIAGRAM	12
G. SOFTWARE PROCESS MODEL	14
H. ARCHITECTURAL PATTERN	16
I. SYSTEM ARCHITECTURE	17
J. DATABASE DESIGN	18
K. USER INTERFACE DESIGN	21
L. USER ACCEPTANCE TEST	26
M. CERTIFICATION	29
N. REFERENCES	30
APPENDIX A. RELEASE READINESS CHECKLIST	31
APPENDIX B. IMPLEMENTATION ISSUES	32
APPENDIX C. SOFTWARE TESTING RESULTS	33
APPENDIX D. USER MANUAL	34
APPENDIX E. DEPLOYMENT AND HANDOFF	38
APPENDIX F. PROJECT RESPONSIBILITIES	39

A. BUSINESS REQUIREMENTS

Business Profile

The TGMCI Radiology Queueing System supports the Radiology Department of Tagum Global Medical Center Inc. It coordinates queue tickets for X-Ray, Ultrasound, and CT Scan and provides a shared view for reception personnel, radiology staff, administrators, and patients waiting for service. Its purpose is to organize service flow and make the current queue visible across connected workstations.

Problem Statement

Radiology queue coordination requires staff to know which tickets are waiting, which patients have been called, and which procedures have been completed. Separate or manually communicated queue records make this coordination harder when several rooms are operating. A common system is needed to issue consistent queue numbers, apply the configured category priority, and reflect service changes on a public display.

Organizational Needs

- A shared operational queue for reception and radiology staff across the local network.
- Consistent procedure selection and classification of in-patient (IPD) and out-patient (OPD) tickets.
- A patient-facing monitor showing queue codes and service status without requiring a patient account.
- Historical ticket summaries and exports for operational review.
- A controlled way to display announcements and media alongside queue information.

A. BUSINESS REQUIREMENTS continued

Objectives of the System

- Generate a procedure-specific queue number after reception selects a procedure and patient category.
- Select IPD tickets before OPD tickets within a procedure and use first-in, first-out ordering within each category.
- Allow authorized radiology staff to call a waiting ticket and complete an active service.
- Refresh staff and public queue views from a common server-side source.
- Retain issue, call, and completion events and provide date-filtered reports.
- Support announcement images, videos, and online media links with playback controls.

Scope and Limitations

The scope covers staff login, queue registration and printing, waiting and serving lists, completion, reports and Excel export, advertisement management, and the public display. The interface provides two X-Ray slots, two Ultrasound slots, and one CT Scan slot. MySQL stores operational records; media files and JSON state support announcement playback.

Clinical orders, diagnostic images, interpretation reports, billing, electronic medical records, appointment booking, and patient self-registration are outside this scope. The system does not establish clinical urgency. IPD and OPD priority is a configured queue rule that hospital personnel must validate for operational use.

This version contains implementation gaps described in Appendix B. A successful interface demonstration or playback unit test does not establish readiness for clinical operations. Network availability, server configuration, printer setup, and browser media permissions affect operation.

B. USER REQUIREMENTS

User Roles and Expectations

Table 1. User roles

Role	Responsibility
Receptionist	Select procedure and patient category, generate and print tickets, review reports, and manage permitted announcements.
Radiology staff	Operate the queue controls for calling and completing tickets. Room-specific accounts receive interface restrictions based on their names.
Administrator	Access reception, queue management, analytics, and advertisement functions.
Waiting patient	Read the public monitor and respond when the ticket number is called. No patient login is required.

User Needs and Expectations

Reception personnel need clear selections and immediate confirmation of the issued number. Radiology personnel need a current waiting list, recognizable room slots, and an unambiguous completion action. Administrators need a consolidated view and traceable records. Patients need readable codes and a clear distinction between incoming and currently served tickets.

The intended role model must be enforced by server checks as well as interface visibility. The current queue endpoints check staff roles, while specific room assignments are represented in the interface. The release tests must verify these distinctions.

B. USER REQUIREMENTS continued

Table 2. Reception and public-display requirements

ID	Category	Description
UR R1	Authentication	Sign in with an assigned staff username and password.
UR R2	Ticket issue	Choose X-Ray, Ultrasound, or CT Scan and IPD or OPD before generating a number.
UR R3	Printing	View the latest ticket and print a queue slip through the browser print dialog.
UR R4	Monitoring	See current queue status from the server and review operational reports.
UR R5	Reporting	Filter reports by date and procedure and download an Excel report.
UR P1	Visibility	Read incoming ticket codes and Now Serving columns on the public monitor.
UR P2	Announcements	View announcements while waiting without signing in.

Table 3. Radiology and administrator requirements

ID	Category	Description
UR S1	Queue control	Call the next eligible ticket for a procedure and mark active service complete.
UR S2	Room view	Identify the appropriate X-Ray, Ultrasound, or CT Scan serving slot.
UR A1	Oversight	Access staff workflow screens and review queue totals.
UR A2	Media	Add or delete ads, set duration and activity, and issue playback commands.
UR A3	Operations	Maintain staff accounts, backups, and release configuration through authorized technical procedures.

C. SYSTEM REQUIREMENTS SPECIFICATION

C.1 Functional Requirements

Table 4. Authentication and queue functions

ID	Requirement	Acceptance criterion
FR 01	Authenticate staff using stored password hashes.	Valid credentials establish a session; invalid credentials do not.
FR 02	Restrict ticket creation to receptionists and administrators.	Other roles receive a denied response.
FR 03	Create a ticket from procedure and category selections.	A waiting ticket and an issued event are stored together.
FR 04	Allocate numbers using a daily procedure counter.	Concurrent creation and date rollover do not produce ambiguous identifiers.
FR 05	Select eligible waiting tickets using category rank and issue order.	IPD precedes OPD; older tickets precede newer tickets in the same category.
FR 06	Allow radiology staff and administrators to call tickets.	The selected ticket becomes serving and the call is recorded.
FR 07	Complete a currently serving ticket.	The completion timestamp and event are saved in one transaction.
FR 08	Provide a shared queue feed.	Waiting, serving, and completed groups match persisted records.

Requirements describe the intended accepted behavior. FR 04 requires correction of the schema and code mismatch before it can be accepted. Attribution of the issued event also needs correction to use the signed-in staff member.

C. SYSTEM REQUIREMENTS continued

Table 5. Display reporting and media functions

ID	Requirement	Acceptance criterion
FR 09	Display incoming and serving queue codes.	Public monitor updates after staff actions without a patient login.
FR 10	Print the issued ticket.	Slip shows the issued code, procedure, patient category, and issue details.
FR 11	Retrieve tickets by start and end dates.	Report results fall inside the selected range.
FR 12	Export the filtered report to Excel.	Workbook values reconcile with the selected report view.
FR 13	Manage advertisement metadata and files.	An authenticated staff user can add and delete supported content.
FR 14	Store playback commands and settings.	Public display receives applicable mode and command changes.
FR 15	Synchronize public-display playback and preview.	The preview follows reported media position without unnecessary restarts.
FR 16	Record operational events.	Issue, call, and completion events reference the affected ticket and actor.

Workflow Boundaries

Waiting to Serving and Serving to Completed are exposed by the current routes. Although the schema includes skipped and cancelled values, the checked-in route set does not provide skip or cancellation actions. Staff account maintenance and a comprehensive audit administration interface are not implemented as dashboard modules.

C.2 NON FUNCTIONAL REQUIREMENTS

Table 6. Quality requirements and validation methods

ID	Category	Requirement and validation
NFR 01	Responsiveness	Target a queue change becoming visible within five seconds on the hospital LAN. Measure end-to-end; the code polls the queue every two seconds.
NFR 02	Usability	Staff should complete ticket selection and generation without ambiguous controls. Observe representative staff tasks during UAT.
NFR 03	Access control	Use password hashes, authenticated staff sessions, and server-side role and room authorization. Test unauthorized requests directly.
NFR 04	Integrity	Store related ticket and event changes transactionally. Test concurrent requests and interruption recovery.
NFR 05	Privacy	Keep names, diagnoses, and clinical identifiers out of the public queue payload and display. Inspect responses as well as screens.
NFR 06	Compatibility	Validate staff screens, public display, printing, and media playback on the hospital browsers and monitors.
NFR 07	Recovery	Back up MySQL and uploaded media and demonstrate a restore to a separate environment before release.
NFR 08	Maintainability	Keep presentation, request handling, and persistence responsibilities identifiable and document schema changes.

The targets above are acceptance criteria, not measured performance results. Browser autoplay restrictions and external media availability must be included in validation. Local queue operation and online announcements have different network dependencies.

D. ACTIVITY DIAGRAM

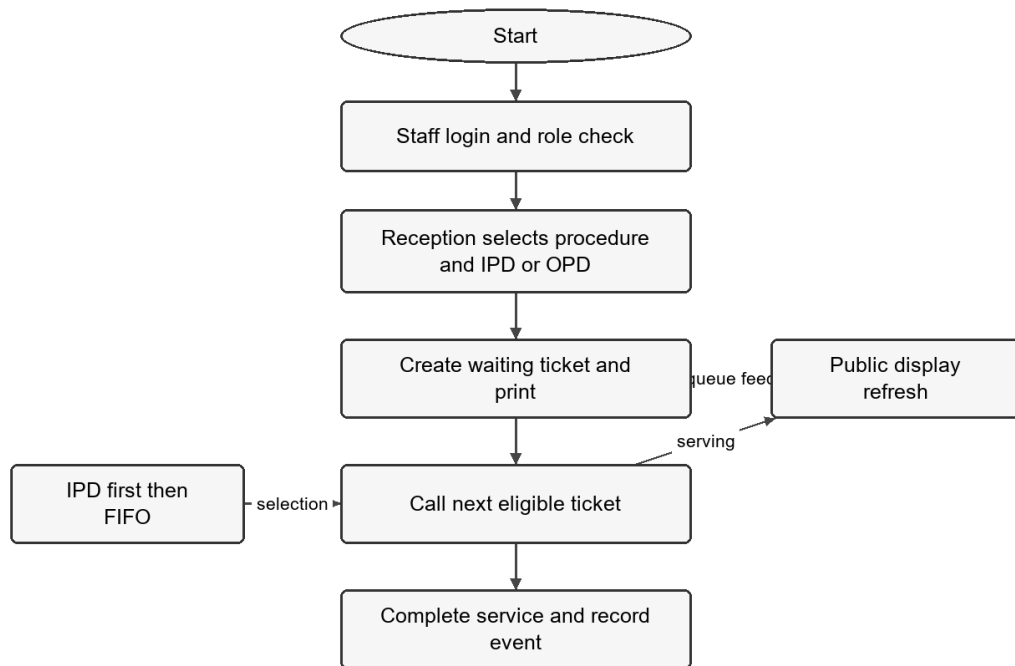


Figure 1. Ticket issue call and completion workflow

Reception selects the procedure and patient category and issues the ticket. Radiology staff call the next eligible ticket when a room is ready, perform the service, and record completion. The public display receives refreshed queue state throughout the workflow. Printing is a browser action following successful ticket creation.

D. ACTIVITY DIAGRAM continued

Main Activity Sequence

Table 7. Activity inputs and outcomes

Step	Action	Outcome
1	Staff sign in.	Session identifies the staff role.
2	Reception selects procedure and IPD or OPD.	Generation becomes available in the interface.
3	Reception requests a number.	Server allocates the sequence and records a waiting ticket.
4	Reception prints the slip.	Patient receives a visible queue code.
5	Staff select Call next for a room.	Server chooses the highest-priority waiting ticket for that procedure.
6	Staff select Complete after service.	Ticket becomes completed and the room can serve another ticket.
7	Display and reports refresh.	Queue views reflect the stored changes.

Alternative and Failure Paths

Missing selections keep the generation button disabled. An unauthorized API request is rejected. An empty queue returns a no-waiting-ticket result. A completion request for a non-serving code returns an error status. A database exception rolls back the active transaction. If printing is blocked, staff must confirm the ticket was already issued before attempting another generation.

Network loss must not be interpreted as successful issue or completion. Staff should reconcile the current queue before retrying a request whose response was interrupted. A documented fallback process is required for server or power failure.

E. USE CASE DIAGRAM

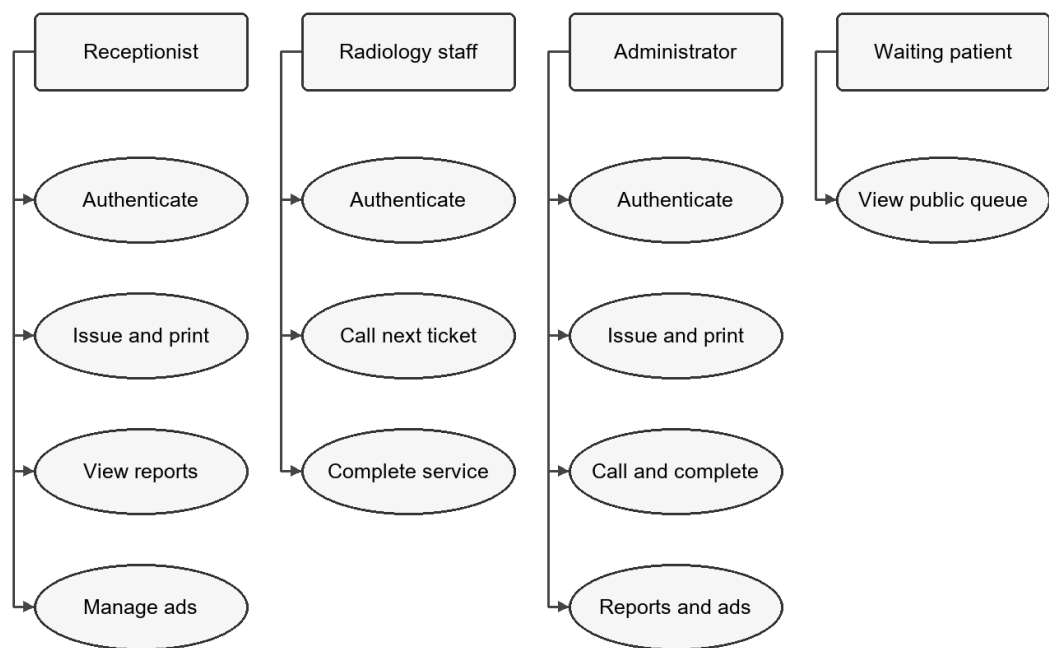


Figure 2. Primary actor interactions

The staff actors share authentication. Reception and administrators issue tickets, while radiology staff and administrators call and complete tickets. The public monitor presents queue information to waiting patients. Account administration and backup operations are technical responsibilities outside the current dashboard use cases.

E. USE CASE DESCRIPTIONS

Table 8. Core use case specifications

Use case	Precondition	Result and exception
UC 01 Staff login	An assigned staff account exists.	Session is created on successful password verification; invalid credentials remain on login.
UC 02 Issue ticket	Receptionist or administrator is signed in; reference records exist.	Waiting ticket and issued event are committed. Invalid input or database errors prevent normal success.
UC 03 Call next	Radiology staff or administrator is signed in; a waiting ticket exists.	Selected ticket becomes serving in the requested slot. Empty queue produces no selected ticket.
UC 04 Complete ticket	A serving ticket exists.	Ticket becomes completed and receives completion time and event.
UC 05 View report	Staff session is present.	Date-filtered generated and completed records are returned.
UC 06 Manage ads	Staff session is present.	Metadata and media are added or deleted; invalid upload conditions return errors.
UC 07 View public queue	Monitor can reach the application.	Incoming and Now Serving information refreshes periodically.

Room-specific authorization is an acceptance requirement beyond the current role check. The public display itself is read-only for queue transitions, but the playback reporting endpoint currently accepts playback-state writes without a staff session.

F. CONTEXT FLOW DIAGRAM

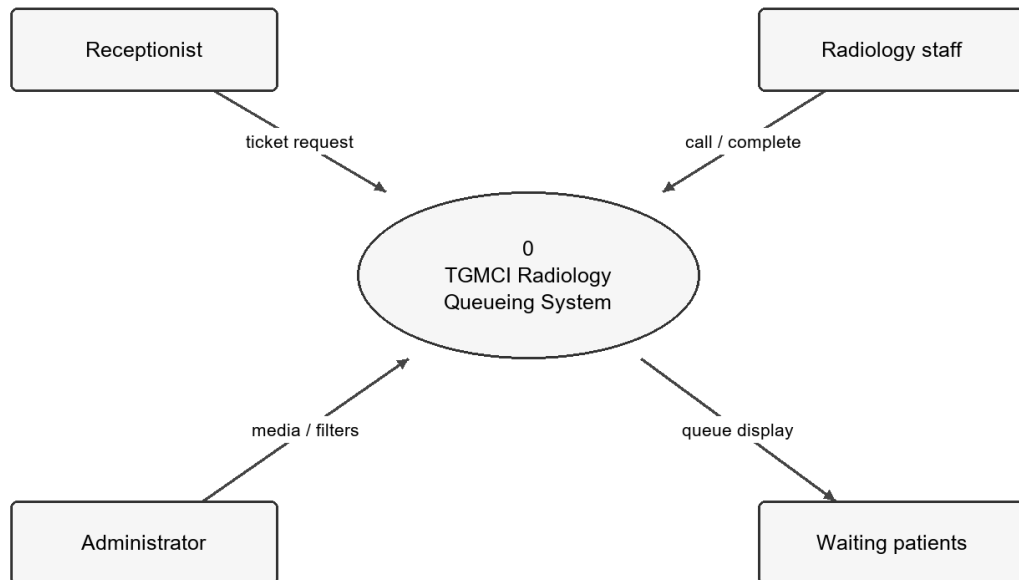


Figure 3. System boundary and external actors

Reception supplies ticket selections and receives the issued code and current queue information. Radiology staff supply call and completion commands and receive updated service state. Administrators provide media and report filters and receive metadata, status, and summaries. Waiting patients receive the rendered public queue and announcements.

MySQL and local media storage are internal components of this system boundary. Online media providers are optional external content sources; they are not responsible for queue selection or persistence.

F. LEVEL 1 DATA FLOW DIAGRAM

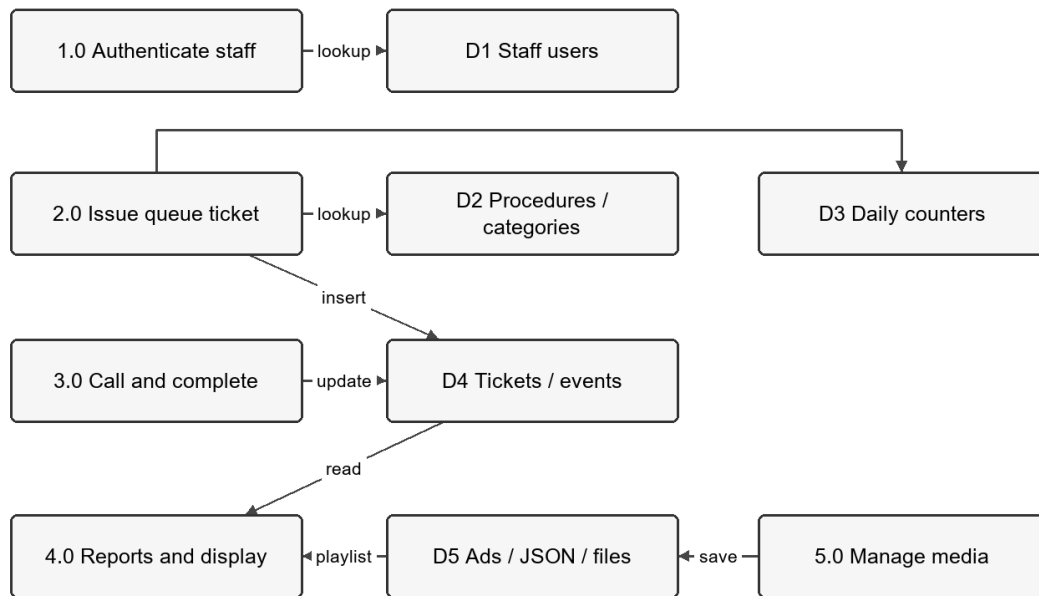


Figure 4. Processes and persistent data stores

Authentication reads staff users. Ticket issuance reads procedures and categories, updates a daily counter, and writes a ticket and event. Queue control updates ticket state and appends events. Reports and the public display read the resulting data. Advertisement management stores metadata in MySQL, content in files, and playback coordination in JSON state. Each arrow identifies a primary data dependency rather than every response message.

G. SOFTWARE PROCESS MODEL

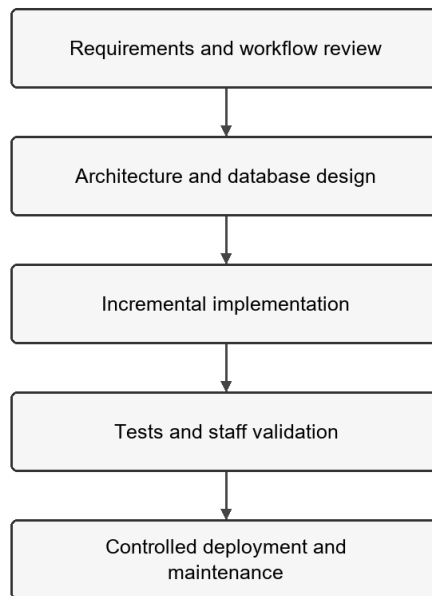


Figure 5. Proposed development and release process

An iterative development process is appropriate for this application because the repository contains successive designs and implementation changes for authentication, queue handling, reporting, and media playback. Each increment should refine a defined workflow, update the schema when needed, and validate the result before the next release. The diagram defines a repeatable process for continuing the project; it does not assert that unrecorded interviews or acceptance activities have occurred.

G. IMPLEMENTATION TECHNOLOGIES

Table 9. Technology and role

Technology	Application role
PHP 8.1 or newer	Request handling, session authentication, routing, services, and repositories. Composer declares PHP >=8.1.
HTML CSS JavaScript	Staff dashboard, public display, printing, polling, media control, and report presentation.
MySQL with PDO MySQL	Relational persistence and prepared queries. SQL scripts specify InnoDB and utf8mb4.
Custom router and autoloader	App\ classes are loaded from app/. Routes map HTTP paths to controller methods.
SheetJS browser library	Creates the downloadable Excel report; loaded from a CDN in the dashboard.
Local files and JSON	Uploaded advertisement content and playback commands, settings, and position.
Git and GitHub	Version history and repository collaboration.
Node test runner	Executes the existing JavaScript playback tests.

Development Phases

Requirements analysis defines actors and priority rules. Design identifies data relationships, API boundaries, and interface states. Implementation adds coherent workflow increments. Testing includes automated logic checks, database integration, cross-workstation behavior, and staff UAT. Deployment follows a restore rehearsal and signed acceptance. Maintenance tracks defects, backups, and changes in browser or media behavior.

H. ARCHITECTURAL PATTERN

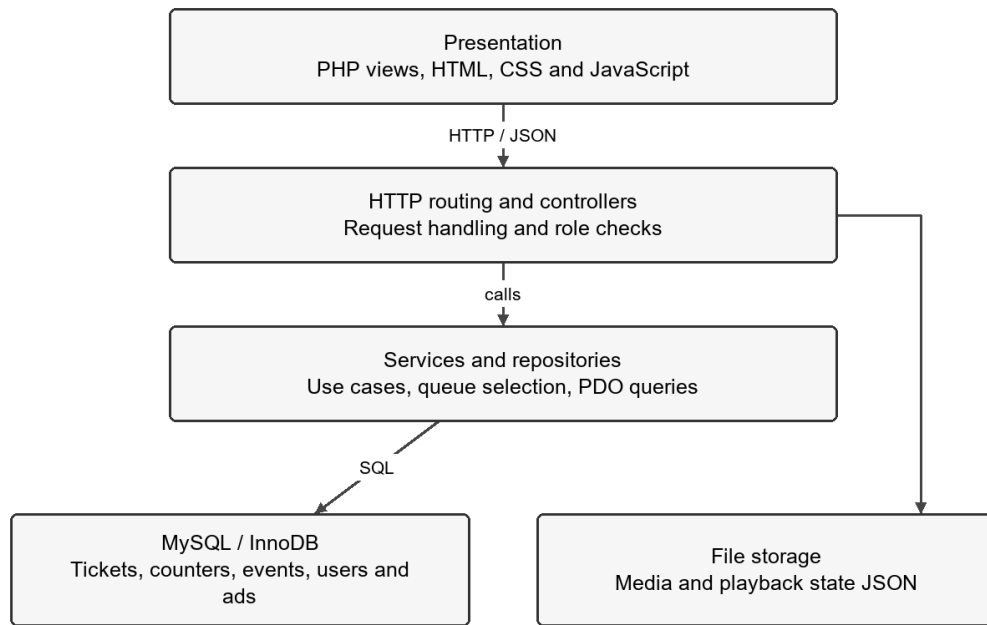


Figure 6. Layered PHP application structure

The application is a modular monolith with presentation, controller, service, and repository responsibilities. Views render PHP templates with client-side JavaScript. The router dispatches requests to controllers. Services support authentication, procedures, and ticket creation, while repositories implement persistence and some queue business logic. Controllers also perform direct file operations for media state. This is a layered design, with some responsibilities still coupled across layers.

I. SYSTEM ARCHITECTURE

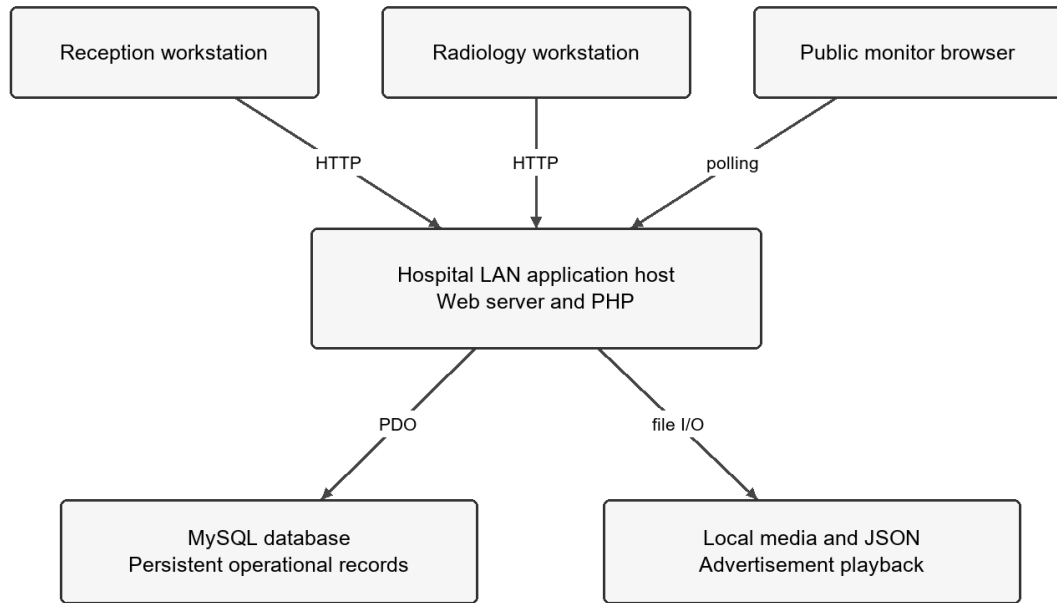


Figure 7. Target local network deployment

Staff workstations and the public monitor connect to one PHP application host on the hospital LAN. The web server document root is public/. The application accesses MySQL through PDO and reads or writes local media and JSON state. The PHP bootstrap sets Asia/Manila, and the database connection sets the session offset to +08:00. Online media and CDN resources require separate internet connectivity.

The architecture depicts the intended deployment arrangement. Server capacity, local URL, HTTPS configuration, backup location, and installation acceptance must be recorded for the actual hospital environment.

J. DATABASE DESIGN

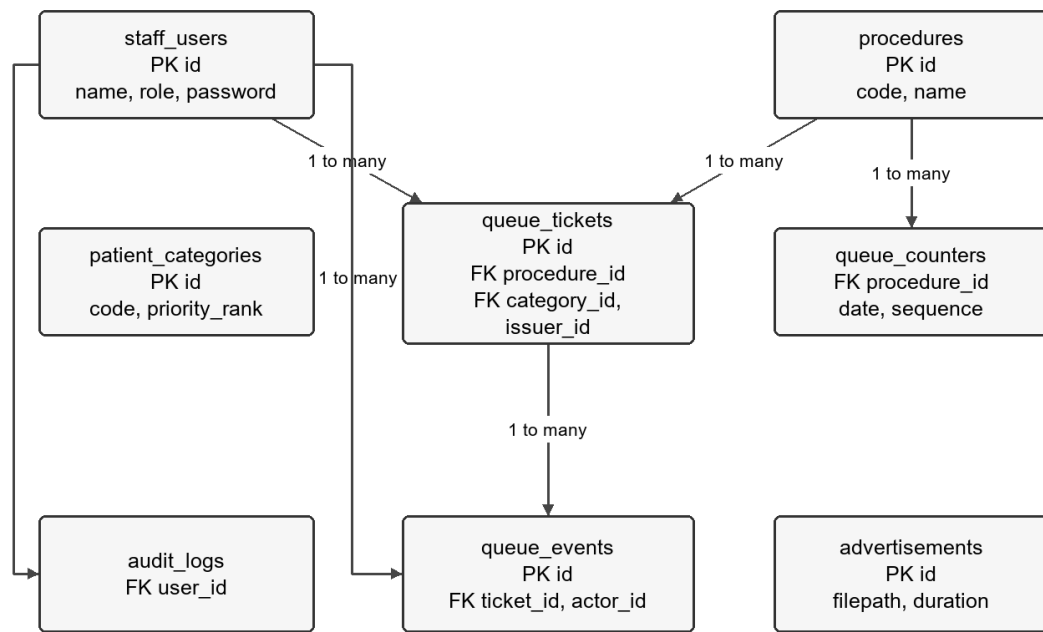


Figure 8. Relationships declared in the SQL schema

The checked-in schema contains eight tables. Procedures and categories classify tickets; staff users issue tickets and act on queue events. Daily counters allocate procedure sequences. Advertisements store media metadata. Audit logs reference staff users but do not declare an advertisement foreign key. Queue events also reference staff_users through actor_id; this additional relationship is described here to keep the diagram readable.

J. DATA DICTIONARY

Table 10. Reference and counter tables

Table	Key fields	Purpose
staff_users	id INT PK; name VARCHAR(255); role ENUM; password VARCHAR(255); timestamps	Staff identity and password hash. Roles are receptionist, radiology_staff, administrator.
procedures	id INT PK; code VARCHAR(50) UNIQUE; name VARCHAR(255); timestamps	Seed values XRAY, UT, and CT identify the supported procedures.
patient_categories	id INT PK; code VARCHAR(50) UNIQUE; priority_rank INT; timestamps	IPD rank 1 precedes OPD rank 2.
queue_counters	id INT PK; date DATE; procedure_id FK; last_sequence_number INT	Unique combination of date and procedure_id tracks daily numbering.

Table 11. Ticket fields

Field group	Definition
Identity	id INT primary key; ticket_code VARCHAR(100) declared globally unique.
Classification	procedure_id and category_id reference the corresponding lookup records; issuer_id references staff_users.
Lifecycle	status ENUM waiting, serving, completed, skipped, cancelled; issue_time defaults to current time; serving_time and completed_time are nullable.
Metadata	created_at and updated_at timestamps. The runtime additionally expects serving_slot, which is absent from the supplied SQL.

J. DATA INTEGRITY AND STORAGE

Table 12. Event advertisement and audit tables

Table	Fields and constraints
queue_events	id PK; ticket_id FK; actor_id FK; action VARCHAR(255); created_at TIMESTAMP. Records issued, called, and completed actions in the implemented flows.
advertisements	id PK; filepath VARCHAR(255); duration_seconds INT default 10; is_active BOOLEAN default true; display_order INT default 0; timestamps.
audit_logs	id PK; user_id FK; action VARCHAR(255); details TEXT; created_at TIMESTAMP. No comprehensive audit-writing workflow is shown by current controllers.

Transaction and Identifier Rules

Ticket creation locks the applicable daily counter, updates or inserts the sequence, creates the ticket, and records the issued event within a transaction. Queue call selects a waiting ticket with FOR UPDATE, then saves the serving state and event. Completion locks a serving ticket and commits its new state and event together.

The generated code uses a prefix and sequence such as XR-001. Because daily counters reset while ticket_code is globally unique, a fresh installation from the supplied SQL can encounter a next-day collision. The release design must choose date-scoped uniqueness or a date-bearing identifier and update code and tests consistently. This documentation does not apply a database migration.

File Storage

Uploaded advertisements are stored under public/storage/uploads/ads/. The storage directory holds ad_command.json, ad_settings.json, and ad_playback.json. Backups must account for both database metadata and the corresponding media files. Playback state is transient operational state, not a substitute for queue history.

K. USER INTERFACE DESIGN

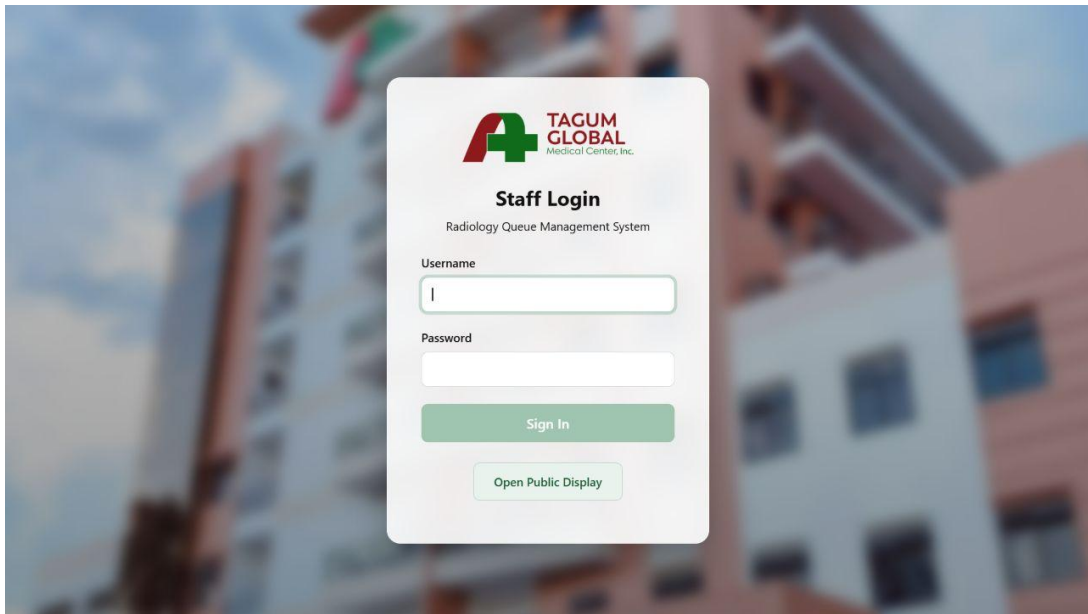


Figure 9. Staff login interface

Login and Navigation

The login screen uses the hospital logo, a background image, username and password fields, a Sign In button, and an Open Public Display link. After login, the dashboard navigation depends on the role. Receptionists and administrators can access reception and analytics; radiology staff receive the queue-management view.

Figures 9 through 15 show actual PHP templates rendered in an isolated documentation preview with empty synthetic API responses. They illustrate the interface and do not constitute production screenshots or proof of database integration.

K. RECEPTION AND QUEUE INTERFACES

The screenshot shows the 'Reception' interface of the 'RADIOLOGY QUEUE MANAGEMENT SYSTEM'. The top header includes the TAGUM GLOBAL Medical Center, Inc. logo, the user 'LOGGED IN AS Documentation Demo Administrator', and the date/time 'THU, SEP 17 11:15:04 AM'. A left sidebar contains navigation links: Dashboard, Reception (highlighted), Manage queue, and Ads. At the bottom of the sidebar are 'Sign Out' and 'Open public display' buttons. The main content area is titled 'Reception' and features a 'Register Patient Queue' section with 'PROCEDURE CATEGORY' (X-Ray, Ultrasound, CT Scan) and 'PATIENT CATEGORY' (In-patient (IPD), Out-patient (OPD)) buttons. A green 'Generate queue number' button is below these. A message states: 'Select a procedure and patient category to continue.' To the right, a 'LATEST TICKET' box shows 'No ticket generated yet this session.'

Figure 10. Reception ticket selection interface

Reception groups procedure and patient category choices before enabling Generate queue number. The Latest Ticket area confirms a successful issue and supports the printing workflow.

The screenshot shows the 'Manage queue' interface. The top header is identical to Figure 10. The left sidebar is the same, but 'Manage queue' is highlighted. The main content area is titled 'Manage queue' and displays three columns for 'X-Ray', 'Ultrasound', and 'CT Scan'. Each column has a status bar at the top (e.g., '0 waiting - 0/2 slots' for X-Ray). Below each status bar are two rows of controls. For X-Ray, the first row (XRAY 1) and second row (XRAY 2) each have a green 'No patient being served' box, a 'Complete' button with a checkmark, and a 'Call next' button with a circular arrow. The 'Ultrasound' and 'CT Scan' columns follow a similar pattern. At the bottom of each column is a 'WAITING LIST' box showing 'Queue is empty'. The right sidebar contains 'Sign Out' and 'Open public display' buttons.

Figure 11. Queue management and serving slots

The queue view groups controls by procedure. Empty queues show inactive Call next controls; each active room provides a Complete action for its serving ticket.

K. REPORTING AND MEDIA INTERFACES

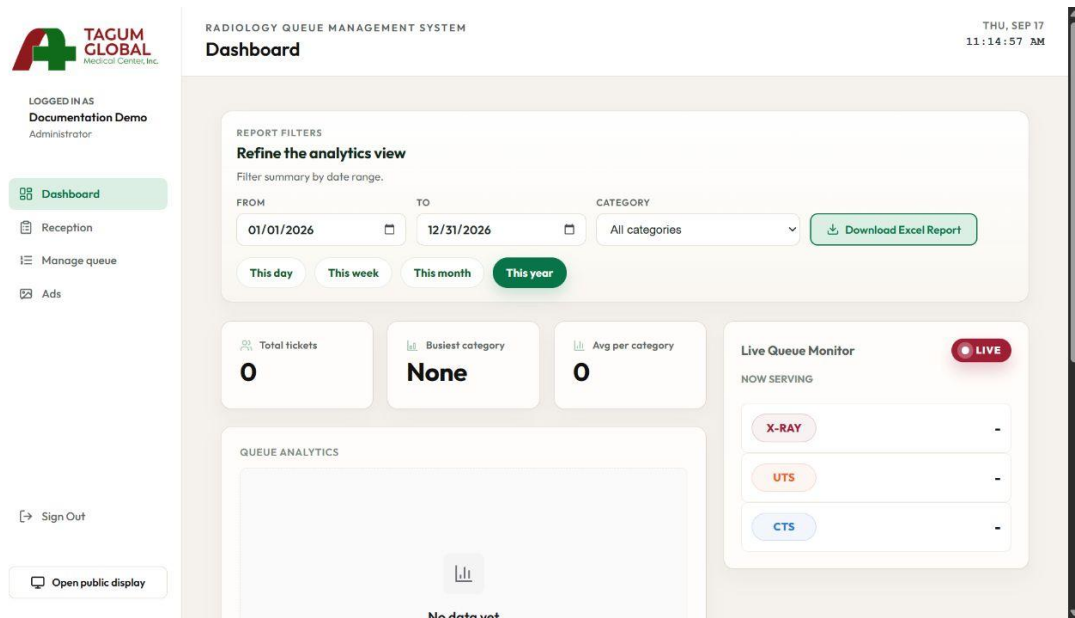


Figure 12. Analytics filters and live queue monitor

The dashboard combines date and category filters, ticket totals, charts, a live monitor, and Download Excel Report.

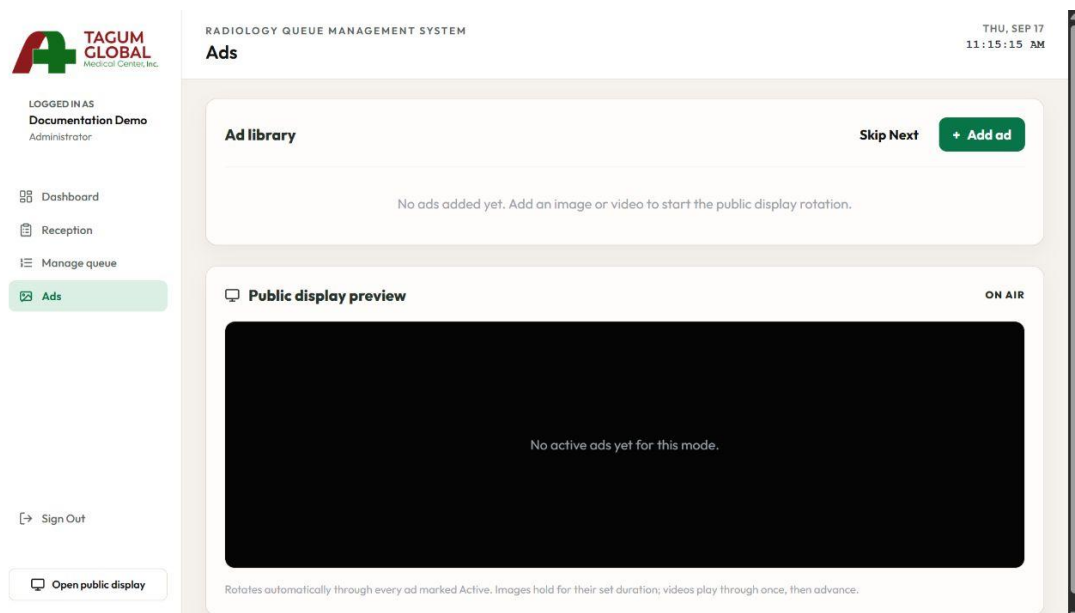


Figure 13. Advertisement library and display preview

The Ads screen includes an ad library, Add ad, Skip Next, and a public-display preview. Uploaded media and external video links are managed through the add form.

K. ROOM SPECIFIC RADIOLOGY STAFF INTERFACE

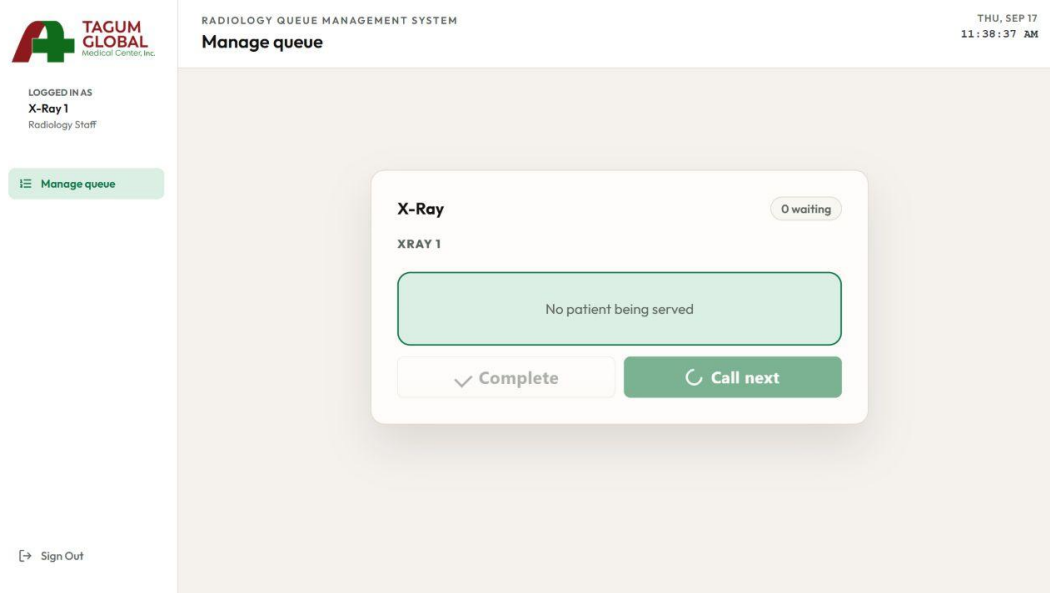


Figure 14. X-Ray 1 room-specific Manage queue interface

When a named room account signs in, the system opens Manage queue and displays only the procedure and serving slot assigned to that username. The room card contains two queue actions: Complete and Call next. Complete is enabled only while that room is serving a ticket, and Call next is enabled only when the room is idle and its procedure has a waiting ticket. The detailed waiting list is hidden for room-specific accounts.

Room account interface mapping

Login account	Visible room card	Queue actions
X-Ray 1	X-Ray slot 1	Complete and Call next
X-Ray 2	X-Ray slot 2	Complete and Call next
Ultrasound 1	Ultrasound slot 1	Complete and Call next
Ultrasound 2	Ultrasound slot 2	Complete and Call next
CT Scan	CT Scan slot 1	Complete and Call next

Room-account navigation shows Manage queue and Sign Out. Dashboard, Reception, Ads, Open public display, other procedure cards, other room slots, and the detailed waiting list are not displayed. The general Radiology User and the administrator can see all five serving slots.

Implementation boundary. The verified restriction is enforced by the dashboard interface using the exact stored username. The queue API currently authorizes the radiology_staff role but does not independently verify the account-to-room assignment. Server-side room authorization is therefore required before the interface restriction can be treated as a security control.

K. PUBLIC DISPLAY INTERFACE

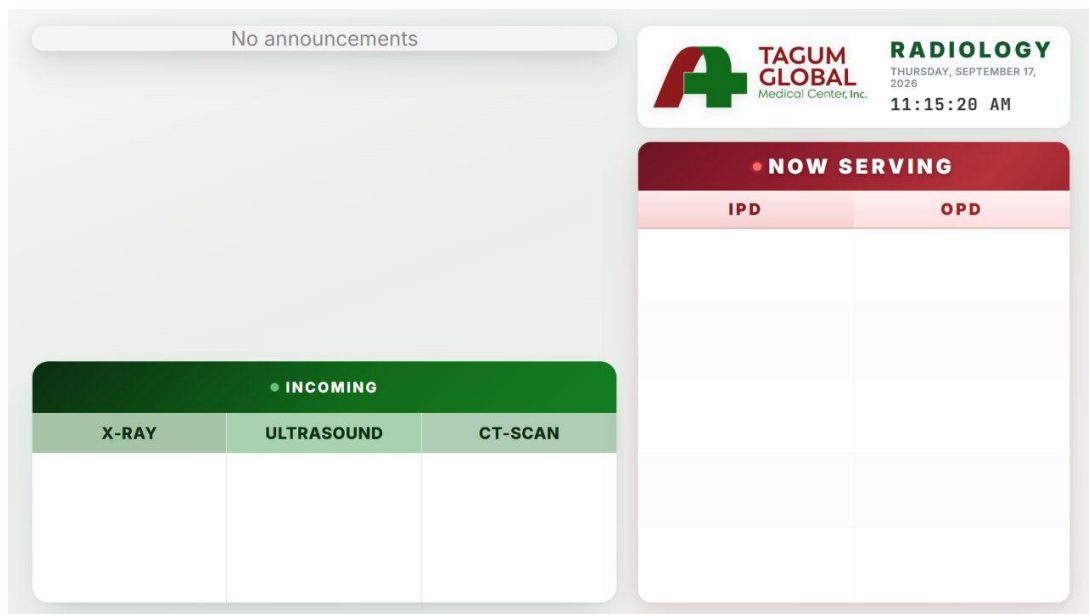


Figure 15. Public queue monitor

Display Content

The monitor combines announcements, incoming tickets by procedure, hospital branding, a clock, and Now Serving columns for IPD and OPD. It uses large text and distinct areas to support viewing at a distance. With no active media, the announcement area displays an empty-state message.

Refresh and Playback

The public queue refresh interval is two seconds. Advertisement polling is configured at 500 milliseconds, and playback reporting at 750 milliseconds. These are configured timer intervals rather than guaranteed response times. Actual latency depends on the browser, network, and server. Staff should verify audio, visibility, and playback on the installed monitor.

L. USER ACCEPTANCE TEST

Purpose and Test Preparation

User acceptance testing verifies that representative reception personnel, radiology staff, and administrators can perform the intended workflow on the hospital equipment. Use a separate test database and clearly marked test tickets. Record the application revision, schema version, browser, printer, monitor, and network environment before execution.

The following scenarios are prepared for execution. Their status is Pending until a tester records an observed result and supporting evidence. They must not be interpreted as successful hospital acceptance results.

Table 13. Authentication and ticket acceptance scenarios

ID	Test procedure	Expected result
UAT 01	Sign in with valid and invalid credentials.	Only valid credentials open the staff dashboard.
UAT 02	Attempt ticket creation as each role and without a session.	Only receptionists and administrators can issue tickets.
UAT 03	Issue one ticket for each procedure and IPD/OPD combination.	Correct code, classification, waiting state, and event are saved.
UAT 04	Issue tickets concurrently from two reception workstations.	No lost issue or duplicated number; failures are clearly reported.
UAT 05	Repeat the first issue after a date rollover.	The new daily number is valid and historical tickets remain distinguishable.
UAT 06	Print the generated ticket.	Slip remains legible and matches the saved ticket.

L. USER ACCEPTANCE TEST continued

Table 14. Queue report and display scenarios

ID	Test procedure	Expected result
UAT 07	Queue OPD then IPD tickets for the same procedure.	IPD is called first; FIFO is preserved within each category.
UAT 08	Call and complete tickets in all five room slots.	The correct slot and lifecycle timestamps are persisted.
UAT 09	Use simultaneous calls and attempt an unauthorized room operation.	No ticket is assigned twice and room policy is enforced server-side.
UAT 10	Observe the public display after issue, call, and completion.	Displayed codes match the server within the agreed latency target.
UAT 11	Select report dates and procedure; download Excel.	The screen and workbook reconcile with test records.
UAT 12	Add image/video content and an online media link.	Applicable content appears with its settings and deletion removes it from rotation.
UAT 13	Change dashboard tabs during playback, pause manually, and return.	Background recovery preserves progress; a manual pause remains paused.
UAT 14	Interrupt the network and restore it.	Staff can reconcile state without duplicating actions.
UAT 15	Restore a backup into a separate environment.	Ticket history, users, and media references are recovered.

L. ACCEPTANCE RECORD

Table 15. Test execution record

Record item	Value to be completed
Test case ID	_____
Tester and role	_____
Execution date	_____
Application and schema revision	_____
Observed result	_____ _____
Evidence location	_____
Outcome	Pass / Fail / Blocked
Defect and retest reference	_____

Acceptance Criteria

The application is eligible for acceptance when required scenarios pass, known blocking defects are corrected, report totals reconcile, authorization checks succeed, and the hospital representatives approve the operational workflow. A failed test must identify the actual observation, reproduction steps, responsible owner, and retest evidence.

Acceptance Decision

Decision: Accept / Accept subject to recorded conditions / Reject

Conditions and unresolved items:

Reception representative: _____ Date: _____

Radiology representative: _____ Date: _____

Administrator or project owner: _____ Date: _____

M. CERTIFICATION

System Review and Acceptance Form

Project: TGMCI Radiology Queueing System

Organization: Tagum Global Medical Center Inc

Department: Radiology

Application revision: _____ Review date: _____

The authorized reviewers shall complete this form after examining the system, test records, user guidance, and deployment requirements. Signing this form records the decision and any conditions stated below.

Review decision:

Approved scope:

Conditions or exclusions:

Authorized hospital representative

Name: _____

Signature: _____ Date: _____

Radiology department representative

Name: _____

Signature: _____ Date: _____

Project representative

Name: _____

Signature: _____ Date: _____

N. REFERENCES

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APPENDIX A. RELEASE READINESS CHECKLIST

Table A1. Release controls

Check	Evidence required
Schema consistency	Versioned migration includes every runtime field, including serving_slot, and resolves daily numbering.
Access and attribution	Role and room checks pass; ticket events identify the signed-in issuer.
Queue correctness	Priority, FIFO, concurrency, occupied-slot, and date-rollover tests pass.
Public display	Payload contains intended queue fields only; monitor and browser behavior are validated.
Media and exports	Upload checks, playback behavior, and Excel results pass on deployed clients.
Operational recovery	Database and media backup restore is demonstrated and recorded.
Staff readiness	UAT results and the accepted manual are available to staff.
Handoff	Server ownership, support contacts, URL, credentials handoff, and acceptance are recorded.

Release date: _____ Responsible owner: _____

Evidence folder or register:

APPENDIX B. IMPLEMENTATION ISSUES

Table B1. Issues identified by source review

Issue	Evidence and required resolution
Missing serving slot	QueueRepository reads and writes serving_slot, but the checked-in SQL omits it. Add a reviewed migration and verify a clean installation.
Daily code collision	The counter resets per date, while ticket_code is globally unique and contains no date. Align identifier generation, lookup, and uniqueness rules.
Issuer attribution	TicketController omits the session user ID when calling TicketService; the default issuer is 1. Pass and validate the authenticated issuer.
Room authorization	QueueController checks radiology_staff but does not enforce the UI room assignment. Enforce assignments on the server.
Slot occupancy	callNext accepts a slot index without an occupied-slot guard. Validate slots and concurrency at the persistence boundary.
Playback writes	The playback endpoint accepts writes without authentication. Define and enforce the intended display-writer authorization model.
Upload validation	Upload handling checks size but relies on the client filename extension. Validate allowed content types, extensions, and URL sources.
Audit coverage	Queue events exist, but broad staff/media audit logging is not implemented. Define required actions and connect them to audit persistence.

These are source-review findings, not claims about an inspected live database. Operational fixes require implementation and retesting before release approval.

APPENDIX C. SOFTWARE TESTING RESULTS

Automated Playback Tests

Execution date: September 17, 2026. Reviewed baseline: 4e024c3. Test runner: bundled Node.js using `node --test` against both repository test files. Observed outcome: 10 tests passed, 0 failed.

Table C1. Executed test coverage

Test group	Observed result
Background HTML video	Five tests passed: uninterrupted tab changes, background pause recovery, manual pause preservation, blocked-playback retry, and replaced/ended video handling.
Background YouTube	One test passed: background recovery with foreground pause behavior.
Playback synchronization	Four tests passed: advancing a playing position, retaining a paused position, negative clock-age handling, and correcting material drift.

Scope of the Results

These unit tests exercise JavaScript behavior with test doubles. They do not verify the MySQL schema, HTTP authentication, ticket numbering, physical printing, real YouTube availability, hospital LAN capacity, or acceptance by users. The documentation preview used empty synthetic API responses and verifies interface layout only.

Integration and User Acceptance Results

Database integration: Pending. Cross-workstation queue tests: Pending. Hospital staff UAT: Pending. Backup restore: Pending. Production deployment certification: Pending.

APPENDIX D. USER MANUAL

1. Getting Started

Open the application URL provided by the system administrator. Use the designated staff workstation for queue operations and the public monitor for patient-facing information. Confirm that the workstation can reach the application and that the printer is available before issuing tickets.

2. Staff Login

- Enter your assigned Username and Password on Staff Login.
- Select Sign In. The system opens the screen allowed for your role.
- If access fails, recheck the username and contact the designated administrator.
Do not use another staff member's account.
- Use Sign Out when leaving the workstation.

3. Navigation

Table D1. Staff navigation

Area	Use
Dashboard	Review date and procedure-filtered reports, summary charts, and the live monitor.
Reception	Choose procedure and patient category and generate a queue number.
Manage queue	Call the next patient and complete service in the appropriate room.
Ads	Manage announcements and observe the display preview.
Open public display	Open the patient-facing monitor in a separate browser tab.

APPENDIX D. TICKET AND QUEUE OPERATIONS

4. Generate and Print a Ticket

- Open Reception.
- Choose X-Ray, Ultrasound, or CT Scan under Procedure Category.
- Choose In-patient (IPD) or Out-patient (OPD) under Patient Category.
- Select Generate queue number after checking both selections.
- Review Latest Ticket and complete the browser print dialog.
- Give the patient the printed ticket and direct them to watch the public monitor.

If a print window does not appear, check the browser popup and printer configuration. A print failure does not necessarily mean ticket creation failed. Confirm the latest saved ticket before generating another number.

5. Call the Next Ticket

- Open Manage queue and locate your assigned procedure and room.
- Check that the room is ready and is not already serving another ticket.
- Select Call next. The system selects from the waiting queue using IPD-first priority and FIFO within a category.
- Verify the displayed serving number and direct the patient to the correct room.

Room-specific screen. X-Ray 1, X-Ray 2, Ultrasound 1, Ultrasound 2, and CT Scan accounts see only their assigned room card. Their available queue actions are Complete and Call next; they do not see the other procedure cards or the detailed waiting list.

6. Complete Service

- Confirm that the displayed ticket is the patient whose service has finished.
- Select Complete for that room and respond to any confirmation dialog.
- Verify that the room no longer shows the completed ticket before calling the next one.

The current user interface has no supported skip or cancellation action. Use the department's authorized exception procedure and record the issue for the administrator.

APPENDIX D. REPORTS AND ANNOUNCEMENTS

7. View and Export Reports

- Open Dashboard and set the From and To dates, or select a date shortcut.
- Select a procedure under Category or retain All categories.
- Review ticket totals, charts, and the live queue monitor.
- Select Download Excel Report and open the downloaded workbook.
- Check the report period and totals before using or distributing the report.

Excel export depends on the browser report library loading successfully. If the interface reports that the library is still loading, restore the required connection or ask the administrator to check the deployment. A file download by itself does not verify report accuracy.

8. Manage Advertisements

- Open Ads and select Add ad.
- Choose the supported upload or online video option shown in the form.
- Set the applicable duration and activity options and submit the content.
- Check the ad library and public-display preview.
- Use Skip Next to advance the display when required; review content before deleting it.

Images use a configured holding duration; videos normally play through before rotation advances. Uploaded files are subject to an application size limit of 50 MB and may also be limited by PHP upload settings. Online video requires connectivity and can be affected by provider or browser playback restrictions.

APPENDIX D. DISPLAY AND TROUBLESHOOTING

9. Operate the Public Monitor

- Open the public-display link from login or the dashboard.
- Place the browser on the designated monitor and use the approved display mode.
- Verify that Incoming and Now Serving are visible at the waiting-area viewing distance.
- Verify audio and announcement playback when audio is required.
- Keep the monitor connected and recheck it after network or power interruptions.

Table D2. Common operating issues

Symptom	Action
No queue number generated	Confirm both selections. If the request failed or timed out, reconcile the latest queue state before retrying.
No ticket printout	Check popup permissions, printer selection, paper, and connectivity. Avoid issuing a duplicate solely to print.
Call next is unavailable	Check whether the waiting list is empty or your interface restricts that room. Escalate inconsistent state.
Monitor appears stale	Check network and server availability, then reload and compare with the staff view.
Announcement will not play	Check active content, browser media permission, file availability, and online connectivity where needed.
Service unavailable	Follow the department-approved manual queue fallback and record tickets for later reconciliation.

Support contact: _____ Local application URL:

APPENDIX E. DEPLOYMENT AND HANDOFF

Installation Sequence

- Prepare the local application host with PHP 8.1 or newer, PDO MySQL, and an appropriate MySQL service.
- Place the project in its approved directory and point the web server document root at public/.
- Configure database environment variables through the PHP process. The bootstrap does not itself load a .env file.
- Reconcile and apply the reviewed schema to a new test database before production setup. Do not import initialization SQL over existing operational data.
- Provision staff accounts through an authorized technical process and replace demonstration credentials.
- Make required media and JSON directories writable by the application account.
- Complete integration tests, backup restore, staff UAT, and the acceptance forms.

Configuration and Rename Impact

The application name is TGMCI Radiology Queueing System. The requested folder and repository name is TGMCI-Radiology-Queueing-System. Renaming the project does not rename the configured database: the existing default remains tgc_radiology_queue. Update any virtual-host document-root path or shortcuts after a successful folder rename. A repository owner must complete any GitHub rename that the connected account cannot perform.

Handoff Evidence Register

Table E1. Deployment evidence

Item	Reference
Host and application URL	_____
Schema and application release	_____
Backup and restore record	_____
UAT and acceptance forms	_____
Support owner and handoff date	_____

APPENDIX F. PROJECT RESPONSIBILITIES

Table F1. Responsibility register

Responsibility	Assigned person
Project coordination and stakeholder review	_____
Requirements and workflow analysis	_____
Interface design and frontend implementation	_____
PHP services and API implementation	_____
Database schema and migration management	_____
Testing and defect tracking	_____
Deployment backup and technical support	_____

Complete the responsibility register with the actual project team. Project biographies, photographs, signatures, and academic submission details should be supplied by the respective contributors for the final submission package.

Document Maintenance

Update the requirements, diagrams, database dictionary, screenshots, and test evidence when the implementation changes. Preserve a release-specific copy of the accepted document and record the application and schema versions used for acceptance.